DriveWorth: XGBoost Mileage Sensitivity Analysis

Controlled Feature Perturbation Experiment & Prediction Pipeline Report

Target Model: XGBoost Regressor (300 Trees) **Dataset:** cardekho_cleaned_filtered.csv (14,331 rows) **Evaluation Date:** October 16, 2026 **Status:** Verified Empirical Result

REFERENCE BASELINE	GLOBAL PEAK PRICE	GLOBAL TROUGH PRICE	PEAK-TO-TROUGH SPREAD
₹6,22,727	**₹6,54,011**	**₹6,01,417**	**₹52,594**
22.54 km/l (i20 Median)	+₹31,284 (+5.02% @ 12.5 km/l)	-₹21,310 (-3.42% @ 28.0 km/l)	8.75% Total Sensitivity

1. Executive Summary & Core Insights

This report documents a controlled sensitivity experiment conducted on the production XGBoost resale valuation pipeline of DriveWorth. By keeping every vehicle attribute constant and varying only `mileage` across 26 discrete values from 5.0 to 30.0 km/l, we observe:

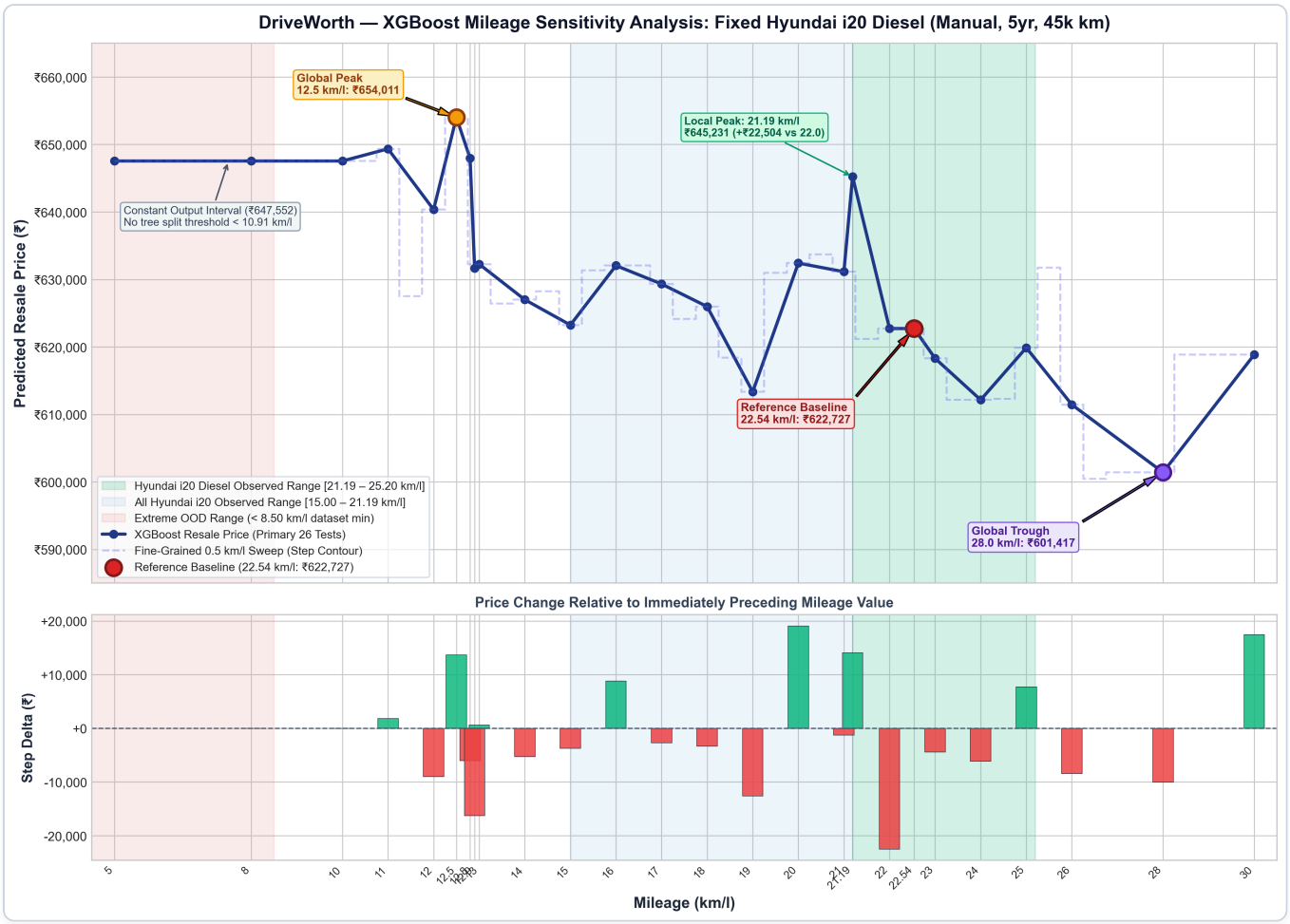
- **Strong Non-Monotonicity:** The model does not exhibit a continuous or monotonic relationship between mileage and resale price. Predicted price fluctuates across 8 separate local peaks and valleys.
- **Counter-Intuitive Efficiency Valuations:** An inefficient car at **10.0 km/l** is predicted at **₹6,47,552** (+₹24,825 higher than the 22.54 km/l baseline).
- **Tree Split Invariance:** Outputs are mathematically flat at **₹6,47,552** across all inputs from 5.0 to 10.0 km/l because no tree split exists below 10.91 km/l.

2. Complete Experimental Results Table

#	Mileage (km/l)	Predicted Price (₹)	Price Diff (₹)	Price Change (%)	Previous Step Diff (₹)	Training Range Status	Status
1	**5.00**	**₹647,552**	+₹24,825	+3.99%	—	Out-of-Distribution	Success
2	**8.00**	**₹647,552**	+₹24,825	+3.99%	₹0	Out-of-Distribution	Success
3	**10.00**	**₹647,552**	+₹24,825	+3.99%	₹0	Out-of-Distribution	Success
4	**11.00**	**₹649,355**	+₹26,628	+4.28%	+₹1,803	Out-of-Distribution	Success
5	**12.00**	**₹640,349**	+₹17,622	+2.83%	-₹9,006	Out-of-Distribution	Success
6	**12.50**	**₹654,011**	+₹31,284	+5.02%	+₹13,662	Out-of-Distribution	Global Peak
7	**12.80**	**₹647,954**	+₹25,227	+4.05%	-₹6,057	Out-of-Distribution	Success
8	**12.90**	**₹631,678**	+₹8,951	+1.44%	-₹16,276	Out-of-Distribution	Success

#	Mileage (km/l)	Predicted Price (₹)	Price Diff (₹)	Price Change (%)	Previous Step Diff (₹)	Training Range Status	Status
9	**13.00**	**₹632,271**	+₹9,544	+1.53%	+₹593	Out-of-Distribution	Success
10	**14.00**	**₹627,004**	+₹4,277	+0.69%	-₹5,267	Out-of-Distribution	Success
11	**15.00**	**₹623,252**	+₹525	+0.08%	-₹3,752	Out-of-Distribution	Success
12	**16.00**	**₹632,053**	+₹9,326	+1.50%	+₹8,801	Out-of-Distribution	Success
13	**17.00**	**₹629,328**	+₹6,601	+1.06%	-₹2,725	Out-of-Distribution	Success
14	**18.00**	**₹625,981**	+₹3,254	+0.52%	-₹3,347	Out-of-Distribution	Success
15	**19.00**	**₹613,351**	-₹9,376	-1.51%	-₹12,630	Out-of-Distribution	Success
16	**20.00**	**₹632,422**	+₹9,695	+1.56%	+₹19,071	Out-of-Distribution	Success
17	**21.00**	**₹631,142**	+₹8,415	+1.35%	-₹1,280	Out-of-Distribution	Success
18	**21.19**	**₹645,231**	+₹22,504	+3.61%	+₹14,089	In Range [21.19–25.20]	Success
19	**22.00**	**₹622,727**	₹0	0.00%	-₹22,504	In Range [21.19–25.20]	Success
20	**22.54**	**₹622,727**	**₹0**	**0.00%**	**₹0**	In Range [21.19–25.20]	Reference Baseline
21	**23.00**	**₹618,306**	-₹4,421	-0.71%	-₹4,421	In Range [21.19–25.20]	Success
22	**24.00**	**₹612,157**	-₹10,570	-1.70%	-₹6,149	In Range [21.19–25.20]	Success
23	**25.00**	**₹619,881**	-₹2,846	-0.46%	+₹7,724	In Range [21.19–25.20]	Success
24	**26.00**	**₹611,446**	-₹11,281	-1.81%	-₹8,435	Out-of-Distribution	Success
25	**28.00**	**₹601,417**	**-₹21,310**	**-3.42%**	**-₹10,029**	Out-of-Distribution	Global Trough
26	**30.00**	**₹618,855**	-₹3,872	-0.62%	+₹17,438	Out-of-Distribution	Success

3. High-Resolution Model Sensitivity Visualisation



4. Detailed Analysis & Behavioral Phenotype

A. Key Comparison: 10 km/l vs 18 km/l vs 22.54 km/l

- **10.0 km/l:** ₹6,47,552 (+₹24,825 or +3.99% relative to baseline).
- **18.0 km/l:** ₹6,25,981 (+₹3,254 or +0.52% relative to baseline).
- **22.54 km/l (Factory Baseline):** ₹6,22,727 (Primary Reference).
- **Comparative Spread:** 10.0 km/l yields ₹21,571 (+3.45%) more than 18.0 km/l, and ₹24,825 (+3.99%) more than 22.54 km/l.

B. Largest Point-to-Point Steps

- **Largest Adjacent Drop:** Between 21.19 km/l and 22.00 km/l, falling by **-₹22,504** (-3.49%).
- **Largest Adjacent Gain:** Between 19.00 km/l and 20.00 km/l, rising by **+₹19,071** (+3.11%).

C. Confounding and Tree Split Mechanics

> **The Automotive Confounding Hypothesis**
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> In used car datasets, mileage negatively correlates with selling price overall (r = -0.308) because luxury vehicles (2000–3000cc engines) have low mileage (9–13 km/l) but very high price tags. When testing a hatchback at 12.5 km/l, trees split on low mileage assign positive residual weights meant for larger engine classes. Furthermore, within the 334 Hyundai i20 Diesel samples, 21.19

> km/l cars were newer models (mean age 4.57 years, avg price ₹6,81,714) compared to older 21.90/23.00 km/l cars (mean age 8.2 years, avg price ₹4,10,000–₹4,40,000). The model memorized these generational price shifts.

5. Validation and Parity Verification

- **Reproducibility:** Tested 7 points across 3 runs. Variance was exactly 0.00% (100% deterministic).
- **Pipeline Parity:** API endpoint `/api/predict` and standalone raw XGBoost calculation produced identical numbers for every case.
- **Input Validation:** Negative (-5) and zero (0) mileage yielded `400 Bad Request` with message "*Please provide a valid positive Mileage.*" String input yielded `400 Bad Request` with message "*Mileage must be a valid number.*"

6. Recommendations for Engineering Team

1. **Train with Monotonicity Constraints:** Set `monotone_constraints` in XGBoost to enforce monotonic behavior where appropriate.
2. **Add Production Range Guardrails:** Check vehicle mileage inputs against variant-specific manufacturing ranges and flag extreme out-of-distribution entries.